

ANALYTICAL METHODS • DENNIS G. ZILL & DR. HAGHIGHAT LECTURE 3

Chapter 2.3: Linear First-Order Differential Equations

Standard canonical form, integrating factor derivation, transient decay, steady-state response, and discontinuous forcing functions.

Zill §2.3 (pp. 79–88)

Lecture 3 Slides 8–15

Guaranteed Midterm Question

PLAIN-ENGLISH INTUITION

When y and y' are linear but inseparable, we collapse the equation into a single derivative using the **Product Rule in reverse**. By multiplying through by the magic multiplier $\mu(x) = e^{\int P(x)dx}$, the left side becomes $\frac{d}{dx}[\mu(x)y]$. Integrating once solves the equation in closed form!

STEP 0: STANDARD CANONICAL FORM (MANDATORY)

Divide by the leading coefficient $a_1(x)$ so the coefficient of y' is exactly 1:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \xrightarrow{\div a_1(x)} \frac{dy}{dx} + P(x)y = f(x)$$

Warning: Failing to divide by $a_1(x)$ destroys the integrating factor!

INTEGRATING FACTOR DERIVATION

Multiply standard form by $\mu(x)$:

$$\mu y' + \mu P(x)y = \frac{d}{dx}[\mu y] = \mu y' + \mu' y.$$

$$\text{Equate } y \text{ coefficients: } \mu' = \mu P(x) \implies \frac{d\mu}{\mu} = P(x)dx.$$

$$\mu(x) = e^{\int P(x)dx}$$

General Solution Formula & Decomposition

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)f(x) dx + C \right] = \underbrace{\frac{C e^{-\int P(x)dx}}{\mu(x)}}_{y_h(x) \text{ (homogeneous)}} + \underbrace{\frac{1}{\mu(x)} \int \mu(x)f(x) dx}_{y_p(x) \text{ (particular)}}$$

Master Worked Problem 1 • Leading Coefficient & Log Signs (Lecture 3 Slide 13)

CLASSIC CONCORDIA EXAM

Problem: Solve the initial value problem: $x \frac{dy}{dx} - 4y = x^5 e^x$, with $y(1) = 0$ for $x > 0$.

1 Standard Form:

Divide by x : $\frac{dy}{dx} - \frac{4}{x}y = x^4 e^x$.

Identify: $P(x) = -\frac{4}{x}$ and $f(x) = x^4 e^x$.

2 Integrating Factor $\mu(x)$:

$$\int P(x)dx = \int -\frac{4}{x}dx = -4 \ln x = \ln(x^{-4}).$$

$$\mu(x) = e^{\ln(x^{-4})} = x^{-4} = \frac{1}{x^4}$$

3 Collapse LHS & Integrate:

$$\frac{d}{dx} [x^{-4}y] = x^{-4}(x^4 e^x) = e^x.$$

$$x^{-4}y = \int e^x dx + C = e^x + C \implies y(x) = x^4 e^x + Cx^4.$$

4 Apply IVP $y(1) = 0$:

$$0 = (1)^4 e^1 + C(1)^4 \implies C = -e.$$

Explicit Solution: $y(x) = x^4(e^x - e)$ on $(0, \infty)$.

2 Piecewise Linear Equations & Transient Analysis

ENGINEERING CONTEXT

In electrical and mechanical circuits, inputs suddenly switch ON or OFF (step functions). The differential equation remains linear, but the right-hand forcing function $f(x)$ is **piecewise defined**. We solve on each interval separately, then adjust integration constants to glue the pieces together smoothly!

Master Worked Example 2 · Discontinuous Forcing Function (Lecture 3 Slide 14 & Zill)

EXAM CHALLENGE

Problem: Solve $\frac{dy}{dx} + y = f(x)$, with $y(0) = 0$, where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$.

1 Interval 1 ($0 \leq x \leq 1$):

$$\frac{dy}{dx} + y = 1 \implies P(x) = 1 \implies \mu(x) = e^x.$$

$$\frac{d}{dx}[e^x y] = e^x \implies e^x y = e^x + c_1 \implies y_1(x) = 1 + c_1 e^{-x}.$$

$$\text{Apply } y(0) = 0 \implies 0 = 1 + c_1 \implies c_1 = -1.$$

$$y_1(x) = 1 - e^{-x} \quad (0 \leq x \leq 1)$$

$$\text{Boundary Value at } x = 1: y_1(1) = 1 - e^{-1}.$$

$$\text{Unified Solution: } y(x) = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases}. \text{ Notice how } y(x) \text{ is continuous everywhere!}$$

2 Interval 2 ($x > 1$):

$$\frac{dy}{dx} + y = 0 \implies \frac{d}{dx}[e^x y] = 0 \implies y_2(x) = c_2 e^{-x}.$$

3 Enforce Continuity at $x = 1$:

$$\lim_{x \rightarrow 1^+} y_2(x) = y_1(1) \implies c_2 e^{-1} = 1 - e^{-1}.$$

$$\text{Multiply by } e: c_2 = e(1 - e^{-1}) = e - 1.$$

$$y_2(x) = (e - 1)e^{-x} \quad (x > 1)$$

TRANSIENT VS STEADY-STATE

- **Transient Term:** Any part of the solution that approaches zero as $x \rightarrow \infty$: $\lim_{x \rightarrow \infty} y_{\text{transient}}(x) = 0$. In the problem above, $(e - 1)e^{-x} \rightarrow 0$ is purely transient.
- **Steady-State:** The persisting response as $x \rightarrow \infty$.

THE TOP 2 PITFALLS IN LINEAR DES

1. **Neglecting to divide by $a_1(x)$:** Computing $\mu(x)$ from an un-normalized equation.
2. **Adding $+C$ after dividing by $\mu(x)$:** Writing $y = \frac{1}{\mu} \int \mu f dx + C$ instead of $\frac{1}{\mu} [\int \mu f dx + C]$.